

First Order Logic - Assignment

1 On Semantics

Exercise 1.1. Let v, v' be valuations on an interpretation $\mathcal{I}$.

- (a) If $v(x) = v'(x)$, for all variables x occurring in a term t then $v(t) = v'(t)$.
- (b) For any closed term t , $v(t) = v'(t)$.

Exercise 1.2.

- (i) Consider $\Gamma := \{\alpha\}$. Then $\Gamma \models \forall x\alpha$. However, $\not\models \alpha \rightarrow \forall x\alpha$.
- (ii) For any interpretation $\mathcal{I}$ and wff α , $\mathcal{I} \models \alpha$, if and only if $\mathcal{I} \models \forall x\alpha$.
In fact, if α^* denotes the *closure* of $\alpha(x_1, \dots, x_n)$, i.e. $\alpha^* := \forall x_1 \dots \forall x_n \alpha$ (fix any order for the quantification), we have
 $\mathcal{I} \models \alpha$, if and only if $\mathcal{I} \models \alpha^*$.
- (iii) All PL axioms are valid in FOL, i.e. e.g. $\models \alpha \rightarrow (\beta \rightarrow \alpha)$, where α, β are FOL-wffs. Similarly for axioms A2, A3 of PL.
- (iv) $\models \forall x(\alpha \rightarrow \beta) \rightarrow (\forall x\alpha \rightarrow \forall x\beta)$, but
 $\not\models (\forall x\alpha \rightarrow \forall x\beta) \rightarrow \forall x(\alpha \rightarrow \beta)$ – find a suitable counterexample. In other words, choose wffs α, β , interpretation $\mathcal{I}$ and valuation v on $\mathcal{I}$ such that $\mathcal{I}, v \models \forall x\alpha \rightarrow \forall x\beta$, but $\mathcal{I}, v \not\models \forall x(\alpha \rightarrow \beta)$.
- (v) $\models \forall x(\alpha \wedge \beta) \leftrightarrow (\forall x\alpha \wedge \forall x\beta)$.
- (vi) $\models \exists x(\alpha \wedge \beta) \rightarrow (\exists x\alpha \wedge \exists x\beta)$, but
 $\not\models (\exists x\alpha \wedge \exists x\beta) \rightarrow \exists x(\alpha \wedge \beta)$. Again, find a suitable counterexample.

Exercise 1.3. Consider any interpretation $\mathcal{I}$, and a FOL-wff α .

(i) Let t be a term with at least one occurrence of the variable x , and s another term. Further, let t' be the term obtained from t by replacing every occurrence of x by the term s .

Let u, v be x -equivalent valuations on $\mathcal{I}$ such that $v(x) = u(s)$.
Then $v(t) = u(t')$.

(ii) ('Extension' of (i) to wffs) Suppose α has at least one free occurrence of the variable x , and let t be a term free for x in α . Let u, v be x -equivalent valuations on $\mathcal{I}$ such that $v(x) = u(t)$.

Then $\mathcal{I}, v \models \alpha(x)$, if and only if $\mathcal{I}, u \models \alpha(t/x)$.

(iii) (Free variables matter) Let u, v be valuations on $\mathcal{I}$ such that $u(x) = v(x)$, for each free variable x in α . Then $\mathcal{I}, u \models \alpha$, if and only if $\mathcal{I}, v \models \alpha$.

(iv) If α is closed, either $\mathcal{I} \models \alpha$, or $\mathcal{I} \models \neg\alpha$.

(v) $\models \forall x\alpha \rightarrow \alpha(t/x)$, if t is free for x in α .

(vi) $\models \alpha \rightarrow \forall x\alpha$, if x is not free in α .

2 Proof theory

2.1 Axiomatisation

'Introduction' and 'elimination' rules for the quantifiers:

Exercise 2.1. The following rules may be derived in *FOL*.

(i) (**Specialization (Spec)**)

$$\frac{\Gamma \vdash \forall x\alpha}{\Gamma \vdash \alpha(t/x)}$$

if the term t is free for x in α .

(ii) (**Existential rule (Rule $\exists$)**)

$$\frac{\Gamma \vdash \alpha(t/x)}{\Gamma \vdash \exists x\alpha}$$

if the term t is free for x in α .

(iii) (**Generalization rule (Gen)**)

$$\frac{\Gamma \vdash \alpha}{\Gamma \vdash \forall x \alpha}$$

if x is not free in Γ (i.e. in any wff of Γ).

(iv) (**Choice rule (Rule C)**)

$$\frac{\Gamma \cup \{\alpha\} \vdash \beta \quad \Gamma \vdash \exists x \alpha}{\Gamma \vdash \beta}$$

if x is not free in Γ , nor in β .

Observe some special cases of the rules.

Corollary 2.1.

(i) From Spec:

$$\frac{\Gamma \vdash \forall x \alpha}{\Gamma \vdash \alpha}$$

Another special case arises, when $\Gamma = \emptyset$.

(ii) From Rule $\exists$:

$$\frac{\Gamma \vdash \alpha}{\Gamma \vdash \exists x \alpha}$$

Again, another special case arises, when $\Gamma = \emptyset$.

(iii) From Gen:

$$\frac{\vdash \alpha}{\vdash \forall x \alpha}$$

(iv) From Rule C:

$$\frac{\{\exists x \alpha\} \cup \{\alpha\} \vdash \beta}{\{\exists x \alpha\} \vdash \beta} \quad \text{if } x \text{ is not free in } \beta.$$

Exercise 2.2. Prove the following.

- (i) $\vdash \forall x(\alpha \rightarrow \beta) \rightarrow (\alpha \rightarrow \forall x\beta)$, whenever x is not free in α .
(As mentioned in the lectures, this is taken as an axiom in some approaches, e.g. in Kelly.)
- (ii) $\vdash \forall x(\alpha \wedge \beta) \leftrightarrow (\forall x\alpha \wedge \forall x\beta)$.
- (iii) $\vdash \exists x(\alpha \wedge \beta) \rightarrow (\exists x\alpha \wedge \exists x\beta)$.
The converse implication is *not* a theorem.
- (iv) $\vdash \forall x(\alpha \rightarrow \beta) \rightarrow (\exists x\alpha \rightarrow \exists x\beta)$.
- (v) $\vdash \exists x(\alpha \rightarrow \beta) \rightarrow (\alpha \rightarrow \exists x\beta)$, x is not free in α .
- (vi) $\vdash \exists x(\alpha \rightarrow \beta) \rightarrow (\forall x\alpha \rightarrow \beta)$, x is not free in β .
- (vii) $\vdash \forall x(\forall y\alpha) \rightarrow \forall y(\forall x\alpha)$.
- (viii) $\vdash \exists x(\exists y\alpha) \rightarrow \exists y(\exists x\alpha)$.

Hint: In order to prove the theorems that involve $\exists$ in the assumption of D.T., one can make use of Rule C – the proofs come very easily.

2.2 Tableaux

Exercise 2.3. Exercises from Chapter 8 in Kelly.

3 First order Theories

3.1 Equality

The equality axioms:

$$\text{E1 } \forall x(x = x).$$

$$\text{E2 } \forall x_1 \dots \forall x_n \forall y_1 \dots \forall y_n (x_1 = y_1 \wedge \dots \wedge x_n = y_n \rightarrow (p^n x_1 \dots x_n \leftrightarrow p^n y_1 \dots y_n)).$$

$$\text{E3 } \forall x_1 \dots \forall x_n \forall y_1 \dots \forall y_n (x_1 = y_1 \wedge \dots \wedge x_n = y_n \rightarrow (f^n x_1 \dots x_n = f^n y_1 \dots y_n)).$$

Exercise 3.1. Using the axioms, prove symmetry and transitivity of $=$, i.e.

- (i) $\forall x \forall y (x = y \rightarrow y = x)$, and

$$(ii) \quad \forall x \forall y \forall z (x = y \wedge y = z \rightarrow x = z).$$

Hint: In axiom E2 above, consider the special case when $n = 2$, and p^2 is $=$.

For any first order theory T with equality, we denote by ‘ $\exists n$ ’, $n \geq 2$, the wff

$$\exists x_1 \dots \exists x_n (x_1 \neq x_2 \wedge x_1 \neq x_3 \wedge \dots \wedge x_1 \neq x_n \wedge x_2 \neq x_3 \wedge \dots \wedge x_{n-1} \neq x_n).$$

Exercise 3.2. Show the following.

- (i) For any interpretation $\mathcal{I}$ of T ,
 $\mathcal{I} \models \exists n$, if and only if $|D| \geq n$,
 where $|D|$ denotes the cardinality of the domain D of $\mathcal{I}$.
- (ii) If $n \geq k$, $\vdash \exists n \rightarrow \exists k$.

‘ $\exists! n$ ’ denotes the wff:

$$\exists x_1 \dots \exists x_n ((x_1 \neq x_2 \wedge x_1 \neq x_3 \wedge \dots \wedge x_1 \neq x_n \wedge x_2 \neq x_3 \wedge \dots \wedge x_{n-1} \neq x_n) \wedge \forall y (y = x_1 \vee y = x_2 \vee \dots \vee y = x_n)).$$

Exercise 3.3.

- (i) Show that for any interpretation $\mathcal{I}$ of T ,
 $\mathcal{I} \models \exists! n$, if and only if $|D| = n$.
- (ii) Express through a wff that “there exist at most two x ’s such that $\alpha(x)$ ”.

3.2 Isomorphism of models

Exercise 3.4.

- (i) Consider two isomorphic models $\mathcal{I}$ and $\mathcal{I}'$ of T , with bijection $h : D \rightarrow D'$. For any wff α of T and valuation v in $\mathcal{I}$,
 $\mathcal{I}, v \models \alpha$, if and only if $\mathcal{I}', h \circ v \models \alpha$.
- (ii) (*Corollary of (i)*) $\mathcal{I} \models \alpha$, if and only if $\mathcal{I}' \models \alpha$.

3.3 Complete and Categorical first order theories

Exercise 3.5.

- (i) AG_2, AG_3 are categorical (and hence complete) theories.
- (ii) AG_4 is not complete (and hence not categorical).
- (iii) Let $\mathcal{I}$ be a model of T . Let the theorems of T' be precisely the wffs of T that are true in $\mathcal{I}$. Show that T' is a consistent and complete extension of T .
- (iv) Prove that if a closed wff α of T is true in every countable model for T then $\vdash_T \alpha$.
- (v) Prove that $\vdash_T \alpha$ if and only if there is a finite set $\{\alpha_1, \dots, \alpha_n\}$ of proper axioms of T such that $(\alpha_1 \wedge \dots \wedge \alpha_n) \rightarrow \alpha$ is valid.
- (vi) Let $\alpha_1, \alpha_2, \dots$ be an infinite sequence of closed wffs of T . Let T' be the result of adjoining all the α_i 's as proper axioms to T . Prove that α is true in every model for T' if and only if there is a finite set $\{\alpha_{i_1}, \dots, \alpha_{i_n}\}$ of α_i 's such that $(\alpha_{i_1} \wedge \dots \wedge \alpha_{i_n}) \rightarrow \alpha$ is true in every model for T .
- (vii) Let T be a first order theory whose proper axioms are an infinite set $\alpha_1, \alpha_2, \dots$ of closed wffs such that $\vdash \alpha_n \rightarrow \alpha_m$, if and only if $n \geq m$. Prove that T must be consistent.
- (viii) Prove that a closed wff α of T is a theorem of T if and only if α is a theorem of every consistent and complete extension of T .
- (ix) Prove that T is complete if and only if every closed wff of T that is true in one countable model for T is true in every countable model for T .